Contents

2 Quantum Fields

Exercise 2.1

The partition function is

$$\begin{aligned} Z &= \sum_n \mathrm{e}^{-\beta\hbar\omega(n+\frac{1}{2})} \\ &= \mathrm{e}^{-\beta\hbar\omega/2}\frac{1}{1-\mathrm{e}^{-\beta\hbar\omega}}\,. \end{aligned} \tag{2.1.1}$$

Let $x = \beta\hbar\omega$,

$$Z = \mathrm{e}^{-x/2}(1-\mathrm{e}^{-x})^{-1}\,. \tag{2.1.2}$$

The internal energy is

$$\begin{aligned} E &= N_A\left(-\frac{\partial \ln Z}{\partial\beta}\right) \\ &= -N_A\hbar\omega\frac{1}{Z}\frac{\partial Z}{\partial x} \\ &= \frac{1}{2}N_A\hbar\omega\left(1+\frac{\mathrm{e}^{-x/2}}{\sinh\frac{x}{2}}\right)\,. \end{aligned} \tag{2.1.3}$$

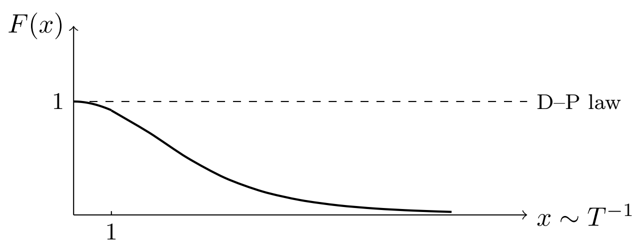
The heat capacity is

$$\begin{aligned} C_V &= \frac{\partial E}{\partial T} = \frac{\mathrm{d}x}{\mathrm{d}T}\frac{\partial E}{\partial x} \\ &= \frac{1}{4}Rx^2\frac{1}{\sinh^2(\frac{x}{2})} \\ &= RF(x) \end{aligned} \tag{2.1.4}$$

where

$$F(x) = \left(\frac{x/2}{\sinh(x/2)}\right)^2 \to 1 \tag{2.1.5}$$

in the $x \to 0$ limit ($k_BT \gg \hbar\omega$), where it recovers the Dulong–Petit law.



Exercise 2.2

$$\begin{aligned} \sum_j \langle q_m|j\rangle\langle j|q_n\rangle &= \frac{1}{N_S}\sum_{j=0}^{N_S-1}\mathrm{e}^{\mathrm{i}(q_n-q_m)ja} \\ &= \frac{1}{N_S}\sum_{j=0}^{N_s-1}\mathrm{e}^{\mathrm{i}(n-m)\frac{2\pi}{N_S}j}\,. \end{aligned} \tag{2.2.1}$$

If $n \neq m$, then[1]

$$\begin{aligned}
\text{LHS} &= \frac{1}{N_S}\frac{\mathrm{e}^{\mathrm{i}2\pi(n-m)}-1}{\mathrm{e}^{\mathrm{i}2\pi(n-m)/N_S}-1} \\
&= \frac{1}{N_S}\frac{\sin(\pi(n-m))}{\sin\left(\frac{\pi}{N_S}(n-m)\right)}\frac{\mathrm{e}^{-\mathrm{i}\pi(n-m)/N_S}}{\mathrm{e}^{-\mathrm{i}\pi(n-m)}} \\
&= \frac{1}{N_S}\frac{\sin(\pi(n-m))}{\sin\left(\frac{\pi}{N_S}(n-m)\right)}(-1)^{\frac{N_S-1}{N_S}(n-m)} \\
&= 0\,.
\end{aligned} \tag{2.2.2}$$

If $n = m$, then

$$\begin{aligned}
\text{LHS} &= \frac{1}{N_S}\sum_{j=0}^{N_S-1} 1 \\
&= 1\,.
\end{aligned} \tag{2.2.3}$$

Therefore,

$$\langle q_m | q_n \rangle = \delta_{mn}\,. \tag{2.2.4}$$

Exercise 2.3

Have

$$\hat{x}(0) = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}+\hat{a}^\dagger) \tag{2.3.1}$$

$$\hat{x}(t) = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}\mathrm{e}^{\mathrm{i}\omega t}+\hat{a}^\dagger\mathrm{e}^{-\mathrm{i}\omega t})\,. \tag{2.3.2}$$

The symmetrised correlation function can be expanded as

$$\begin{aligned}
S_\beta(t) &= \frac{1}{2}\left\langle \hat{x}(t)\hat{x}(0)+\hat{x}(0)\hat{x}(t)\right\rangle \\
&= \frac{\hbar}{4m\omega}\left[2\mathrm{e}^{\mathrm{i}\omega t}\langle\hat{a}\hat{a}\rangle + 2\mathrm{e}^{-\mathrm{i}\omega t}\left\langle\hat{a}^\dagger\hat{a}^\dagger\right\rangle + 2\cos\omega t\left(\left\langle\hat{a}\hat{a}^\dagger\right\rangle+\left\langle\hat{a}^\dagger\hat{a}\right\rangle\right)\right]\,.
\end{aligned} \tag{2.3.3}$$

At (inverse) temperature β,

$$\begin{cases}
\left\langle\hat{a}^\dagger\hat{a}\right\rangle = \langle\hat{n}\rangle \\
\left\langle\hat{a}\hat{a}^\dagger\right\rangle = \left\langle\hat{a}^\dagger\hat{a}+1\right\rangle = \langle\hat{a}\rangle + 1 \\
\langle\hat{a}\hat{a}\rangle = \left\langle\hat{a}^\dagger\hat{a}^\dagger\right\rangle = 0\,,
\end{cases} \tag{2.3.4}$$

where

$$\langle\hat{n}\rangle = n(\beta) = \frac{1}{\mathrm{e}^{\beta\hbar\omega}-1}\,. \tag{2.3.5}$$

Substituting in (2.3.3) gives

$$S_\beta(t) = \frac{\hbar}{2m\omega}\cos(\omega t)\Big[\underbrace{2n(\beta)}_{\text{finite-}T\text{ excitation}} + \underbrace{1}_{\text{zero-point fluctuation}}\Big]\,. \tag{2.3.6}$$

[1] I believe the stated result is missing the red N_S^{-1}.

Exercise 2.4

(a)

$$
\begin{aligned}
[\hat{b}, \hat{b}] &= u^2[\hat{a}, \hat{a}] + uv[\hat{a}, \hat{a}^\dagger] + uv[\hat{a}^\dagger, \hat{a}] + v^2[\hat{a}^\dagger, \hat{a}^\dagger] \\
&= 0 \qquad (2.4.1) \\
[\hat{b}^\dagger, \hat{b}^\dagger] &= u^2[\hat{a}^\dagger, \hat{a}^\dagger] + uv[\hat{a}^\dagger, \hat{a}] + uv[\hat{a}, \hat{a}^\dagger] + v^2[\hat{a}, \hat{a}] \\
&= 0 \qquad (2.4.2) \\
[\hat{b}, \hat{b}^\dagger] &= u^2[\hat{a}, \hat{a}^\dagger] + uv[\hat{a}, \hat{a}] + uv[\hat{a}^\dagger, \hat{a}^\dagger] + v^2[\hat{a}^\dagger, \hat{a}] \\
&= u^2 - v^2 \,. \qquad (2.4.3)
\end{aligned}
$$

The first two canonical commutation relations of $\hat{b}$ are guaranteed by the canonical commutation relations of $\hat{a}$, while the last one satisfies $[\hat{b}, \hat{b}^\dagger] = 1$ if and only if $u^2 - v^2 = 1$.

This means that it is convenient to parameterise u, v as $u = \cosh\theta$, $v = \sinh\theta$.

(b) Given $u^2 - v^2 = 1$, the inverse transform is

$$\hat{a} = u\hat{b} - v\hat{b}^\dagger\hat{a}^\dagger = u\hat{b}^\dagger - v\hat{b}\,. \qquad (2.4.4)$$

Then we can expand

$$
\begin{aligned}
\hat{a}^\dagger\hat{a} + \frac{1}{2} &= u^2\hat{b}^\dagger\hat{b} - uv\hat{b}^\dagger\hat{b}^\dagger - uv\hat{b}\hat{b} + v^2\hat{b}^\dagger\hat{b}^\dagger + \frac{1}{2} \\
&= (u^2 + v^2)\left(\hat{b}^\dagger\hat{b} + \frac{1}{2}\right) - uv(\hat{b}^\dagger\hat{b}^\dagger + \hat{b}\hat{b}) \qquad (2.4.5)
\end{aligned}
$$

and similarly

$$\hat{a}\hat{a} + \hat{a}^\dagger\hat{a}^\dagger = (u^2 + v^2)(\hat{b}\hat{b} + \hat{b}^\dagger\hat{b}^\dagger) - 4uv\left(\hat{b}^\dagger\hat{b} + \frac{1}{2}\right) \qquad (2.4.6)$$

using $u^2 - v^2 = 1$. Therefore,

$$\hat{H} = [\omega(u^2 + v^2) - 2\Delta uv]\left(\hat{b}^\dagger\hat{b} + \frac{1}{2}\right) + \left[\frac{\Delta}{2}(u^2 + v^2) - \omega uv\right](\hat{b}^\dagger\hat{b}^\dagger + \hat{b}\hat{b})\,. \qquad (2.4.7)$$

For $\hat{H}$ to be diagonalisable, there must be some θ such that for $u = \cosh\theta$, $v = \sinh\theta$, the latter term vanish,

$$\Delta(u^2 + v^2) = 2\omega uv\,. \qquad (2.4.8)$$

In θ, this is

$$\Delta\cosh 2\theta = \omega\sinh 2\theta\,, \qquad (2.4.9)$$

so

$$\tanh 2\theta = \frac{\Delta}{\omega}\,. \qquad (2.4.10)$$

Therefore, we must have

$$|\Delta| < \omega \qquad (2.4.11)$$

for $\hat{H}$ to be diagonalisable.

The new frequency is

$$
\begin{aligned}
\tilde{\omega} &= \omega(u^2 + v^2) - 2\Delta uv \\
&= \omega\cosh 2\theta - \Delta\sinh 2\theta\,. \qquad (2.4.12)
\end{aligned}
$$

Given $\tanh 2\theta = \frac{\Delta}{\omega}$, we have

$$\begin{cases} \sinh 2\theta = \frac{\Delta}{\sqrt{\omega^2-\Delta^2}} \\ \cosh 2\theta = \frac{\omega}{\sqrt{\omega^2-\Delta^2}} \, , \end{cases} \tag{2.4.13}$$

so

$$\tilde{\omega} = \sqrt{\omega^2 - \Delta^2} \, . \tag{2.4.14}$$

Since

$$\begin{cases} u^2 + v^2 = \cosh 2\theta = \frac{\omega}{\tilde{\omega}} \\ u^2 - v^2 = 1 \, , \end{cases} \tag{2.4.15}$$

we have

$$\begin{cases} u^2 = \frac{1}{2}\left(\frac{\omega}{\tilde{\omega}} + 1\right) \\ v^2 = \frac{1}{2}\left(\frac{\omega}{\tilde{\omega}} - 1\right) \, . \end{cases} \tag{2.4.16}$$

We need to take care of the signs when taking the square roots.

$$\begin{cases} u = \sqrt{\dfrac{1}{2}\left(\dfrac{\omega}{\sqrt{\omega^2-\Delta^2}} + 1\right)} \\ v = \operatorname{sgn}(\Delta)\sqrt{\dfrac{1}{2}\left(\dfrac{\omega}{\sqrt{\omega^2-\Delta^2}} - 1\right)} \, . \end{cases} \tag{2.4.17}$$

When $\Delta = \omega$, the transition is singular. As $\theta \to \infty$, $u, v \to \infty$ but $\tilde{\omega} \to 0$. However, this does not mean that $\hat{H}$ is 0 (as it does not, apparently). In fact,

$$\begin{aligned} \hat{H} &= \frac{1}{2}\omega(\hat{a} + \hat{a}^\dagger)^2 \\ &= \omega \hat{x}^2 \, . \end{aligned} \tag{2.4.18}$$

The momentum term vanishes.

(c) We have

$$\mathrm{e}^{-\alpha \hat{a}^\dagger \hat{a}^\dagger} = \sum_{k=0}^{\infty} \frac{1}{k!}\left(-\alpha \hat{a}^\dagger \hat{a}^\dagger\right)^k \, . \tag{2.4.19}$$

The commutation relations are

$$[\hat{a}^\dagger, \mathrm{e}^{-\alpha \hat{a}^\dagger \hat{a}^\dagger}] = 0 \tag{2.4.20}$$

and

$$\begin{aligned} [\hat{a}, \mathrm{e}^{-\alpha \hat{a}^\dagger \hat{a}^\dagger}] &= \sum_{k=0}^{\infty} \frac{1}{k!}(-\alpha)^k \left[\hat{a}, (\hat{a}^\dagger)^{2k}\right] \\ &= \sum_{k=1}^{\infty} \frac{1}{k!}(-\alpha)^k \cdot 2k(\hat{a}^\dagger)^{2k-1} \, , \end{aligned} \tag{2.4.21}$$

where in evaluating $[\hat{a}, (\hat{a}^\dagger)^n]$, notice that when you move $\hat{a}$ to the right of a string of $\hat{a}^\dagger$'s, each time you move through an $\hat{a}^\dagger$, you get a $[\hat{a}, \hat{a}^\dagger] = 1$ multiplied with the other $(n-1)$ $\hat{a}^\dagger$'s.

Therefore,

$$
\begin{aligned}
\hat{b}\left|\tilde{0}\right\rangle &= (u\hat{a} + v\hat{a}^\dagger)\mathrm{e}^{-\alpha\hat{a}^\dagger\hat{a}^\dagger}\left|0\right\rangle \\
&= \left[u\left(\mathrm{e}^{-\alpha\hat{a}^\dagger\hat{a}^\dagger}\hat{a} + [\hat{a}, \mathrm{e}^{-\alpha\hat{a}^\dagger\hat{a}^\dagger}]\right) + v\mathrm{e}^{-\alpha\hat{a}^\dagger\hat{a}^\dagger}\hat{a}^\dagger\right]\left|0\right\rangle \\
&= \left\{u\sum_{k=1}^{\infty}\frac{1}{k!}(-\alpha)^k 2k(\hat{a}^\dagger)^{2k-1} + v\sum_{k=0}^{\infty}\frac{1}{k!}(-\alpha)^k(\hat{a}^\dagger)^{2k+1}\right\}\left|0\right\rangle \\
&= \sum_{k=0}^{\infty}\frac{1}{k!}(-\alpha)^k\left[-\frac{\alpha}{k+1}\cdot 2(k+1)u + v\right](\hat{a}^\dagger)^{2k+1}\left|0\right\rangle\ .
\end{aligned}
\tag{2.4.22}
$$

For this to be 0, each term in the sum must be zero, and hence

$$
\alpha = \frac{v}{2u}\,. \tag{2.4.23}
$$

Alternatively: suppose

$$
\left|\tilde{0}\right\rangle = \sum_{n=0}^{\infty} c_n\left|n\right\rangle\ . \tag{2.4.24}
$$

Then

$$
\begin{aligned}
\hat{b}\left|\tilde{0}\right\rangle &= (u\hat{a} + v\hat{a}^\dagger)\sum_{n=0}^{\infty} c_n\left|n\right\rangle \\
&= u\sum_{n=1}^{\infty} c_n\sqrt{n}\left|n-1\right\rangle + v\sum_{n=0}^{\infty} c_n\sqrt{n+1}\left|n+1\right\rangle \\
&= uc_1\left|0\right\rangle + \sum_{n=1}^{\infty}(uc_{n+1}\sqrt{n+1} + vc_{n-1}\sqrt{n})\left|n\right\rangle\ .
\end{aligned}
\tag{2.4.25}
$$

Each term must vanish, so

$$
\begin{cases}
uc_1 = 0 \\
uc_{n+1}\sqrt{n+1} + vc_{n-1}\sqrt{n} = 0 \quad \text{for } n \geq 1\,.
\end{cases}
\tag{2.4.26}
$$

The first condition implies $c_1 = 0$, while the second condition gives the iterative formula for $n \geq 2$:

$$
c_n = -\frac{v}{u}\sqrt{\frac{n-1}{n}}c_{n-2}\,. \tag{2.4.27}
$$

This means that all the odd terms must vanish:

$$
c_{2k+1} = 0 \tag{2.4.28}
$$

for $k \in \mathbb{N}$, while the even coefficients are

$$
c_{2k} = \left(-\frac{v}{u}\right)^k\sqrt{\frac{(2k-1)!!}{(2k)!!}}c_0\,. \tag{2.4.29}
$$

Therefore,

$$\begin{aligned}
|\tilde{0}\rangle &= \sum_{k=0}^{\infty} c_{2k}\,|2k\rangle \\
&= \sum_{k=0}^{2k} \left(-\frac{v}{u}\right)^k \sqrt{\frac{(2k-1)!!}{(2k)!!}}\,\frac{(\hat{a}^\dagger)^{2k}}{(2k)!}\,|0\rangle \\
&= \sum_{k=0} \left(-\frac{v}{u}\right)^k \frac{1}{(2k)!!}(\hat{a}^\dagger)^{2k}\,|0\rangle \\
&= \left\{\sum_{k=0}^{\infty} \frac{1}{k!}\left[-\frac{v}{2u}(\hat{a}^\dagger)^2\right]^k\right\}|0\rangle \\
&= \exp\left(-\frac{v}{2u}\hat{a}^\dagger\hat{a}^\dagger\right)|0\rangle\ .
\end{aligned} \tag{2.4.30}$$

Note that the new vacuum is not the old vacuum. It contains indefinite, even number of the original bosons.

Exercise 2.5

(a) Let $u_\ell = \phi_{2k}$, $v_\ell = \phi_{2k+1}$. The equation of motion is

$$\begin{cases} m\ddot{u}_\ell = -k(2u_\ell - v_{\ell-1} - v_\ell) \\ M\ddot{v}_\ell = -k(2v_\ell - u_\ell - u_{\ell+1})\,. \end{cases} \tag{2.5.1}$$

Transforming into the Fourier basis

$$\begin{cases} U_q = \frac{1}{\sqrt{N}}\sum_q \mathrm{e}^{-2\mathrm{i}\ell a q} u_\ell \\ V_q = \frac{1}{\sqrt{N}}\sum_q \mathrm{e}^{-2\mathrm{i}\ell a q} v_\ell\,, \end{cases} \tag{2.5.2}$$

where q lives in the first Brillouin zone $q \in (-\frac{\pi}{2a}, \frac{\pi}{2a}]$, the equation of motion is

$$\begin{cases} m\ddot{U}_q = -2kU_q + 2k\cos(aq)V_q \\ M\ddot{V}_q = -2kV_q + 2k\cos(aq)U_q\,. \end{cases} \tag{2.5.3}$$

In matrix notation,

$$\begin{pmatrix} m & 0 \\ 0 & M \end{pmatrix}\begin{pmatrix} \ddot{U}_q \\ \ddot{V}_q \end{pmatrix} = -\begin{pmatrix} 2k & -2k\cos(qa) \\ -2k\cos(qa) & 2k \end{pmatrix}\begin{pmatrix} U_q \\ V_q \end{pmatrix}\,, \tag{2.5.4}$$

or

$$\mathsf{M}\ddot{\mathbf{U}}_q = -\mathsf{K}_q\mathbf{U}_q\,. \tag{2.5.5}$$

Using the ansatz $\mathbf{U}_q(t) = \mathbf{A}_q \mathrm{e}^{\mathrm{i}\omega_q t}$, this is

$$\omega_q^2 \mathsf{M}\mathbf{A}_q = \mathsf{K}_q\mathbf{A}_q\,. \tag{2.5.6}$$

This is a generalised eigenvalue equation, which is not the easiest to solve. This inspires us to introduce the mass-scaled coordinates

$$\mathbf{X}_q = \begin{pmatrix} X_q \\ Y_q \end{pmatrix} = \begin{pmatrix} \sqrt{m}U_q \\ \sqrt{M}V_q \end{pmatrix} \tag{2.5.7}$$

such that

$$\mathbf{U}_q = \mathsf{M}^{-1/2}\mathbf{X}_q\,. \tag{2.5.8}$$

Then we have

$$\ddot{\mathbf{X}} = -\mathsf{D}_q \mathbf{X}_q \tag{2.5.9}$$

where

$$\mathsf{D} = \mathsf{M}^{-1/2}\mathsf{K}\mathsf{M}^{-1/2} = \begin{pmatrix} \frac{2k}{m} & -\frac{2k\cos(qa)}{\sqrt{mM}} \\ -\frac{2k\cos(qa)}{\sqrt{mM}} & \frac{2k}{M} \end{pmatrix} . \tag{2.5.10}$$

Now suppose $\mathbf{X}_q(t) = \mathbf{A}_q \mathrm{e}^{\mathrm{i}\omega_q t}$, we have the eigenvalue equation

$$\mathsf{D}_q \mathbf{A}_q = \omega_q^2 \mathbf{A}_q \, . \tag{2.5.11}$$

Solving for the eigenvalue equation

$$\det(\mathsf{D}_q - \omega_q^2 \mathsf{I}) = 0 \tag{2.5.12}$$

gives the two eigenvalues

$$\omega_\pm^2(q) = k\left(\frac{1}{m} + \frac{1}{M}\right) \pm k\sqrt{\left(\frac{1}{m} + \frac{1}{M}\right)^2 - \frac{4}{mM}\sin^2(qa)} \, . \tag{2.5.13}$$

We denote the two normalised eigenvectors

$$\mathbf{e}_\pm(q) = \begin{pmatrix} e_{\pm,1} \\ e_{\pm,2} \end{pmatrix} . \tag{2.5.14}$$

They satisfy

$$\left(\frac{2k}{m} - \omega_\pm^2\right) e_{\pm,1} - \frac{2k\cos(qa)}{\sqrt{mM}} e_{\pm,2} = 0 \, , \tag{2.5.15}$$

which gives

$$\frac{e_{\pm,2}}{e_{\pm,1}} = \frac{\sqrt{mM}\left(\frac{2k}{m} - \omega_\pm^2\right)}{2k\cos(qa)} \, . \tag{2.5.16}$$

This should be solved along with the normalisation condition

$$e_{\pm,1}^2 + e_{\pm,2}^2 = 1 \, , \tag{2.5.17}$$

but I do not bother to.

Let

$$\mathsf{E}_q = \begin{pmatrix} | & | \\ \mathbf{e}_+(q) & \mathbf{e}_-(q) \\ | & | \end{pmatrix} , \tag{2.5.18}$$

then

$$\mathsf{E}_q^{\mathrm{T}} \mathsf{D}_q \mathsf{E}_q = \begin{pmatrix} \omega_+^2(q) & 0 \\ 0 & \omega_-^2(q) \end{pmatrix} =: \Omega_q \, , \tag{2.5.19}$$

and so

$$\ddot{\mathbf{X}}_q = -\mathsf{E}_q \Omega_q \mathsf{E}_q^{\mathrm{T}} \mathbf{X}_q \, . \tag{2.5.20}$$

Therefore

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}(\mathsf{E}_q^{\mathrm{T}} \mathbf{X}_q) = -\Omega_q (\mathsf{E}_q^{\mathrm{T}} \mathbf{X}_q) \, . \tag{2.5.21}$$

This is a diagonal equation of motion, and

$$\mathbf{Q}_q = \mathsf{E}_q^{\mathrm{T}} \mathbf{X}_q \tag{2.5.22}$$

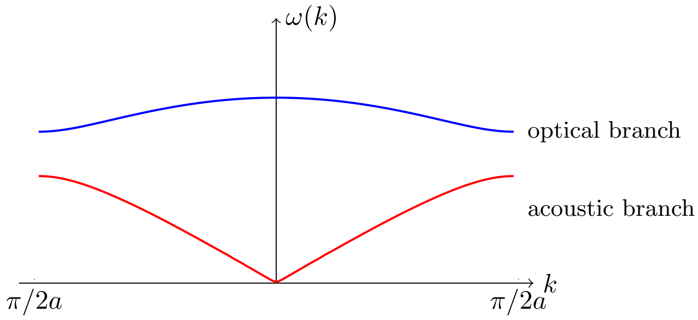
are the normal coordinates. The two normal coordinates are

$$Q_{q,\pm} = \mathbf{e}_{q,\pm}^{\mathrm{T}} \mathbf{X}_q \, . \tag{2.5.23}$$

They satisfy

$$\ddot{Q}_{q,\pm} = -\omega_{q,\pm}^2 Q_{q,\pm} \, , \tag{2.5.24}$$

so that each is oscillating independently with frequency $\omega_{q,\pm}$.



(b) At the band edge $q = \frac{\pi}{2a}$,

$$\omega_\pm^2 = \frac{k}{mM}\left[(m+M) \pm (m-M)\right], \tag{2.5.25}$$

so

$$\begin{cases} \omega_+ = \sqrt{\frac{2k}{M}} \\ \omega_- = \sqrt{\frac{2k}{m}} \end{cases} . \tag{2.5.26}$$

The band gap is

$$\Delta\omega = \sqrt{2k}\left|\frac{1}{\sqrt{M}} - \frac{1}{\sqrt{m}}\right| . \tag{2.5.27}$$

(c) We will denote the momentum conjugate to $\mathbf{X}_q$ as $\mathbf{P}$, and that to $\mathbf{Q}_q$ as $\mathbf{\Pi}_q$. The kinetic energy is

$$\begin{aligned} T &= \frac{1}{2}\sum_q m\left|\dot{U}_q\right|^2 + M\left|\dot{V}_q\right|^2 \\ &= \frac{1}{2}\sum_q \dot{\mathbf{X}}_q^\dagger \dot{\mathbf{X}}_q \\ &= \frac{1}{2}\sum_q \mathbf{P}_q^\dagger \mathbf{P}_q \\ &= \frac{1}{2}\sum_q \mathbf{\Pi}_q^\dagger \mathbf{\Pi}_q , \end{aligned} \tag{2.5.28}$$

and the potential energy is

$$\begin{aligned} V &= \frac{1}{2}\sum_q \mathbf{X}_q^\dagger \mathsf{D}_q \mathbf{X}_q \\ &= \frac{1}{2}\sum_q \mathbf{Q}_q^\dagger \Omega_q \mathbf{Q}_q . \end{aligned} \tag{2.5.29}$$

The Hamiltonian is therefore

$$H = \sum_q \sum_{\sigma=\pm}\left[\frac{1}{2}\left|\Pi_{q,\sigma}\right|^2 + \frac{1}{2}\omega_{q,\sigma}^2\left|Q_{q,\sigma}\right|^2\right] . \tag{2.5.30}$$

Quantising:

$$\hat{a}_{q,\sigma} = \sqrt{\frac{\omega_{q,\sigma}}{2\hbar}}Q_{q,\sigma} + \frac{\mathrm{i}}{\sqrt{2\hbar\omega_{q,\sigma}}}\Pi_{q,\sigma} \tag{2.5.31}$$

such that

$$[\hat{a}_{q,\sigma}, \hat{a}_{q',\sigma'}^\dagger] = \delta_{qq'}\delta_{\sigma\sigma'} . \tag{2.5.32}$$

The Hamiltonian is

$$\hat{H} = \sum_q \sum_{\sigma=\pm}\left[\hbar\omega_{q,\sigma}\left(\hat{a}_{q,\sigma}^\dagger \hat{a}_{q,\sigma} + \frac{1}{2}\right)\right] . \tag{2.5.33}$$

Exercise 2.6

(a) The RMS displacement
$$\Delta r_{\mathrm{RMS}} = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}\,, \tag{2.6.1}$$
where in each direction,
$$\Delta x \sim \sqrt{\frac{\hbar}{m\omega}} \tag{2.6.2}$$
by the uncertainty principle. Therefore, by Linderman criterion, the crystal is unstable if
$$\Delta r_{\mathrm{RMS}} \sim \sqrt{\frac{\hbar}{m\omega}} \gtrsim \frac{1}{3}a \implies \frac{\hbar}{m\omega a^2} > \zeta_c \tag{2.6.3}$$
for some dimensionless ζ_c.

(b) Have
$$\begin{aligned}\hat{\Phi}_j &= \frac{1}{\sqrt{N_S}}\sum_{\mathbf{q}} \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{R}_j}\hat{\Phi}_{\mathbf{q}} \\ &= \sqrt{\frac{\hbar}{2mN_S}}\sum_{\mathbf{q}} \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{R}_j}\frac{1}{\sqrt{\omega_{\mathbf{q}}}}(\hat{a}_{\mathbf{q}} + \hat{a}^\dagger_{-\mathbf{q}})\,.\end{aligned} \tag{2.6.4}$$
Take $j = 0$,
$$\hat{\Phi} = \sqrt{\frac{\hbar}{2mN_S}}\sum_{\mathbf{q}} \frac{1}{\sqrt{\omega_{\mathbf{q}}}}(\hat{a}_{\mathbf{q}} + \hat{a}^\dagger_{-\mathbf{q}})\,. \tag{2.6.5}$$

To work out $\omega_{\mathbf{q}}$, consider
$$\Phi_{\mathbf{R}+\mathbf{a}} - \Phi_{\mathbf{R}} = \frac{1}{\sqrt{N_S}}\sum_{\mathbf{q}} \Phi_{\mathbf{q}}\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{R}}(\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{a}} + 1)\,. \tag{2.6.6}$$
By Fourier orthogonality
$$\sum_{\mathbf{R}} \mathrm{e}^{\mathrm{i}(\mathbf{q}+\mathbf{q}')\cdot\mathbf{R}} = N\delta_{\mathbf{q}',-\mathbf{q}}\,, \tag{2.6.7}$$
we have
$$\begin{aligned}\sum_{\mathbf{R}}(\Phi_{\mathbf{R}+\mathbf{a}} - \Phi_{\mathbf{R}})^2 &= \sum_{\mathbf{q}} \Phi_{\mathbf{q}}\Phi_{-\mathbf{q}}(\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{a}} - 1)(\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{a}} - 1) \\ &= \sum_{\mathbf{q}} 4\sin^2\left(\frac{\mathbf{q}\cdot\mathbf{a}}{2}\right)\Phi^\dagger_{\mathbf{q}}\Phi_{\mathbf{q}}\,.\end{aligned} \tag{2.6.8}$$
Summing over the three directions $\mathbf{a} \in \{a\hat{\mathbf{x}}, a\hat{\mathbf{y}}, a\hat{\mathbf{z}}\}$,
$$V = \frac{1}{2}m\omega^2\sum_{\mathbf{q}} 4\sin\left[\sin^2\frac{q_x a}{2} + \sin^2\frac{q_y a}{2} + \sin^2\frac{q_z a}{2}\right]\Phi^\dagger_{\mathbf{q}}\Phi_{\mathbf{q}}\,, \tag{2.6.9}$$
and hence
$$\omega_{\mathbf{q}} = 2\omega\sqrt{\sin^2\frac{q_x a}{2} + \sin^2\frac{q_y a}{2} + \sin^2\frac{q_z a}{2}}\,. \tag{2.6.10}$$

Therefore,

$$\begin{aligned}\left\langle 0\middle|\hat{\Phi}^2\middle|0\right\rangle &= \frac{\hbar}{2mN_S}\sum_{\mathbf{q},\mathbf{q}'}\frac{1}{\sqrt{\omega_{\mathbf{q}}\omega_{\mathbf{q}'}}}\left\langle 0\middle|(\hat{a}_{\mathbf{q}}+\hat{a}^\dagger_{-\mathbf{q}})(\hat{a}_{\mathbf{q}'}+\hat{a}^\dagger_{-\mathbf{q}'})\middle|0\right\rangle \\ &= \frac{\hbar}{2mN_S}\sum_{\mathbf{q},\mathbf{q}'}\frac{1}{\sqrt{\omega_{\mathbf{q}}\omega_{\mathbf{q}'}}}\underbrace{\left\langle 0\middle|\hat{a}_{\mathbf{q}}\hat{a}^\dagger_{-\mathbf{q}'}\middle|0\right\rangle}_{\delta_{\mathbf{q},-\mathbf{q}'}} \\ &= \frac{\hbar}{2mN_S}\sum_{\mathbf{q}}\frac{1}{\omega_{\mathbf{q}}}\,. \end{aligned} \tag{2.6.11}$$

We will approximate the sum by an integral

$$\frac{1}{N_S}\sum_{\mathbf{q}} \longrightarrow \left(\frac{a}{2\pi}\right)^3\int_{-\pi/a}^{\pi/a}\mathrm{d}^3\mathbf{q}\,. \tag{2.6.12}$$

Then

$$\begin{aligned}\left\langle 0\middle|\hat{\Phi}^2\middle|0\right\rangle &= \frac{\hbar}{4\pi\omega}\int_{-\pi}^{\pi}\frac{-\pi}{\pi}\frac{\mathrm{d}^3\mathbf{x}}{(2\pi)^3}\frac{1}{\sqrt{\sin^2(\frac{x}{2})+\sin^2(\frac{y}{2})+\sin^2(\frac{z}{2})}} \\ &= \frac{\hbar}{4m\omega}I\,,\end{aligned} \tag{2.6.13}$$

where

$$I = 0.910688\ldots \tag{2.6.14}$$

For the crystal to be unstable,

$$\left\langle 0\middle|\hat{\Phi}^2\middle|0\right\rangle > \frac{\hbar}{4m\omega}I > \frac{1}{9}a^2\,, \tag{2.6.15}$$

and so

$$\frac{\hbar}{m\omega a^2} > \frac{4}{9I} =: \zeta_c \approx 0.4880\ldots \tag{2.6.16}$$

(c) Under the unrealistic assumption that ω is independent of a, Helium is *stable* if

$$\frac{\hbar}{m\omega a^2} < \zeta_c\,, \tag{2.6.17}$$

giving

$$a > a_c = \sqrt{\frac{\hbar}{m\omega\zeta_c}} \approx 2.878\times 10^{-11}\ \mathrm{m}\,. \tag{2.6.18}$$

This is apparently unrealistic since crystalline Helium should be *unstable* at low density (for a greater than some a_c).

Using the realistic assumption that $\omega \sim a^{-\alpha}$ for some $\alpha > 2$, we now have that Helium is *stable* if

$$\frac{\hbar}{m\omega a^2} = \frac{\hbar}{m\omega_0}a^\gamma < \zeta_c \tag{2.6.19}$$

for some $\gamma = \alpha - 2 > 0$. This is

$$a < \left(\frac{\zeta_c m\omega_0}{\hbar}\right)^{1/\gamma} =: a_c\,. \tag{2.6.20}$$

Exercise 2.7

(a) Transforming into the Fourier basis,

$$\begin{aligned}\sum_j \hat{a}^\dagger_{j+1}\hat{a}_j &= \sum_j\sum_q\sum_p \mathrm{e}^{-\mathrm{i}(q-p)ja}\mathrm{e}^{-\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}_p \\ &= \sum_q\sum_p \delta_{pq}\mathrm{e}^{-\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}_p \\ &= \sum_q \mathrm{e}^{-\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}_q\,, \end{aligned} \tag{2.7.1}$$

and its Hermitian conjugate is

$$\sum_q \mathrm{e}^{+\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}_q\,. \tag{2.7.2}$$

Therefore, the first term in the Hamiltonian is

$$\hat{H}_q = J_1\sum_q 2\cos(qa)\hat{a}^\dagger_q\hat{a}_q\,. \tag{2.7.3}$$

Next,

$$\begin{aligned}\sum_j \hat{a}^\dagger_{j+1}\hat{a}^\dagger_j &= \sum_j\sum_q\sum_p \mathrm{e}^{-\mathrm{i}(q+p)ja}\mathrm{e}^{-\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}^\dagger_p \\ &= \sum_q\sum_p \delta_{p(-q)}\mathrm{e}^{-\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}^\dagger_p \\ &= \sum_q \mathrm{e}^{-\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}^\dagger_{-q}\,, \end{aligned} \tag{2.7.4}$$

Since $\hat{a}^\dagger_q$ commutes with $\hat{a}^\dagger_{-q}$, we can set $q \to -q$ and write

$$\sum_j \hat{a}^\dagger_{j+1}\hat{a}^\dagger_j = \sum_q \mathrm{e}^{+\mathrm{i}qa}\hat{a}^\dagger_q\hat{a}^\dagger_{-q}\,. \tag{2.7.5}$$

Taking the average of the two expressions,

$$\sum_j \hat{a}^\dagger_{j+1}\hat{a}^\dagger_j = \sum_q \cos(qa)\hat{a}^\dagger_q\hat{a}^\dagger_{-q}\,, \tag{2.7.6}$$

and its Hermitian conjugate is

$$\sum_q \cos(qa)\hat{a}_q\hat{a}_{-q}\,, \tag{2.7.7}$$

so the second part of the Hamiltonian is

$$\hat{H}_2 = J_2\sum_j \cos(qa)(\hat{a}^\dagger_q\hat{a}^\dagger_{-q} + \hat{a}_q\hat{a}_{-q})\,. \tag{2.7.8}$$

The full Hamiltonian is

$$\hat{H} = \sum_q \Big[\underbrace{2J_1\cos(qa)}_{\omega_q}\hat{a}^\dagger_q\hat{a}_q + \underbrace{J_2\cos(qa)}_{\frac{1}{2}\Delta_q}(\hat{a}^\dagger_q\hat{a}^\dagger_{-q} + \hat{a}_q\hat{a}_{-q})\Big] \tag{2.7.9}$$

Now we carry out canonical transformation as we did in Exercise 2.4:

$$\hat{b}_q = u_q\hat{a}_q + v_q\hat{a}^\dagger_{-q}\,. \tag{2.7.10}$$

This is slightly different but you can check that it does transform the Hamiltonian in the same way. Note that the q-dependent factor in ω_q and $\frac{1}{2}\Delta_q$ is identical and hence can be factor out, so if the canonical transformation were to diagonalise $\hat{H}$, the u_q and v_q should be independent of q. We will denote them as $u = \cosh\theta$ and $v = \sinh\theta$.

Now the condition for $\hat{H}$ to be diagonalisable is

$$\tanh 2\theta = \frac{\omega_q}{\Delta_q} = \frac{J_2}{J_1} \in (-1, 1)\,. \tag{2.7.11}$$

This gives

$$\hat{H} = \sum_q \tilde{\omega_q}\left(\hat{b}_q \hat{b}_{-q} + \frac{1}{2}\right)\,, \tag{2.7.12}$$

where

$$\tilde{\omega}_q = \sqrt{\omega_q^2 - \Delta_q^2} = 2\cos(qa)\sqrt{J_2^2 - J_1^2}\,. \tag{2.7.13}$$

As discussed in Exercise 2.4, when $J_1 = J_2$, the transition is singular and

$$\begin{aligned}\hat{H} &= J\sum_j (\hat{a}_{j+1} + \hat{a}_{j+1}^\dagger)(\hat{a}_j + \hat{a}_j^\dagger) \\ &= 2J\sum_j \hat{x}_{j+1}\hat{x}_j\,. \end{aligned} \tag{2.7.14}$$

Exercise 2.8

(a) From Schrödinger equation,

$$\mathrm{i}\hbar\partial_t \mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar}\left|\psi_\mathrm{I}(t)\right\rangle = (\hat{H}_0 + \hat{V}(t))\mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar}\left|\psi_\mathrm{I}(t)\right\rangle\,. \tag{2.8.1}$$

Expanding the derivative,

$$\mathrm{i}\hbar\left[\cancel{-\frac{\mathrm{i}\hat{H}_0}{\hbar}\mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar}} + \mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar}\partial_t\right]\left|\psi_\mathrm{I}(t)\right\rangle = (\cancel{\hat{H}_0} + \hat{V}(t))\mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar}\left|\psi_\mathrm{I}(t)\right\rangle\,, \tag{2.8.2}$$

giving

$$\mathrm{i}\hbar\mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar}\partial_t\left|\psi_\mathrm{I}(t)\right\rangle = \hat{V}(t)\mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar}\left|\psi_\mathrm{I}(t)\right\rangle\,, \tag{2.8.3}$$

and so

$$\mathrm{i}\hbar\partial_t\left|\psi_\mathrm{I}(t)\right\rangle = \hat{V}_\mathrm{I}(t)\left|\psi_\mathrm{I}(t)\right\rangle\,, \tag{2.8.4}$$

where

$$\begin{aligned}\hat{V}_\mathrm{I}(t) &= \mathrm{e}^{+\mathrm{i}\hat{H}_0 t/\hbar}(-f(t)\hat{x})\mathrm{e}^{-\mathrm{i}\hat{H}_0 t/\hbar} \\ &= -f(t)\hat{x}(t)\,. \end{aligned} \tag{2.8.5}$$

(b) Solving

$$\mathrm{i}\hbar\partial_t\left|\psi_\mathrm{I}(t)\right\rangle = \hat{V}_\mathrm{I}(t)\left|\psi_\mathrm{I}(t)\right\rangle \tag{2.8.6}$$

iteratively gives

$$\begin{aligned}\left|\psi_\mathrm{I}(t)\right\rangle &= \left|\psi_\mathrm{I}(-\infty)\right\rangle + \int_{-\infty} \mathrm{d}t'\,\partial_{t'}\left|\psi_\mathrm{I}(t')\right\rangle \\ &= \left|0\right\rangle - \frac{\mathrm{i}}{\hbar}\int_{-\infty}^t \mathrm{d}t'\,\hat{V}_\mathrm{I}(t')\underbrace{\left|\psi_\mathrm{I}(t')\right\rangle}_{\left|0\right\rangle - \frac{\mathrm{i}}{\hbar}\int_{-\infty}^t \mathrm{d}t''\,\hat{V}_\mathrm{I}(t'')\left|\psi_\mathrm{I}(t'')\right\rangle}\,. \end{aligned} \tag{2.8.7}$$

Truncating this series to the $O(\hat{V}_{\mathrm{I}})$ gives

$$\begin{aligned}|\psi_{\mathrm{I}}(t)\rangle &= |0\rangle - \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{t}\mathrm{d}t'\,\hat{V}_{\mathrm{I}}(t')\,|0\rangle + O(\hat{V}_{\mathrm{I}}^2)\\ &= |0\rangle + \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{t}\mathrm{d}t'\,f(t')\hat{x}(t')\,|0\rangle + O(f^2)\,,\end{aligned} \tag{2.8.8}$$

and

$$\langle\psi_{\mathrm{I}}(t)| = \langle 0| - \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{t}\mathrm{d}t'\,f(t')\,\langle 0|\,\hat{x}(t') + O(f^2)\,. \tag{2.8.9}$$

(c)

$$\begin{aligned}\langle\hat{x}(t)\rangle &= \langle\psi_{\mathrm{I}}(t)|\hat{x}(t)|\psi_{\mathrm{I}}(t)\rangle\\ &= \langle 0|\hat{x}(t)|0\rangle - \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{t}\mathrm{d}t'\,f(t')\,\langle 0|\hat{x}(t')\hat{x}(t)|0\rangle + \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{t}\mathrm{d}t''\,f(t'')\,\langle 0|\hat{x}(t)\hat{x}(t'')|0\rangle + O(f^2)\\ &= \underbrace{\langle 0|\hat{x}(t)|0\rangle}_{0} + \frac{\mathrm{i}}{\hbar}\int_{-\infty}^{t}\mathrm{d}t'\,f(t')\,\underbrace{\langle 0|[\hat{x}(t)\hat{x}(t)']|0\rangle}_{R(t-t')} + O(f^2)\\ &= \int_{-\infty}^{t}\mathrm{d}t'\,R(t-t')f(t') + O(f^2)\,.\end{aligned} \tag{2.8.10}$$

If we define $f(t)$ such that it is switched on only after $t=0$, and define $R(\tau)=0$ for $\tau<0$, then we can rewrite

$$\langle\hat{x}(t)\rangle = \int_{0}^{\infty}\mathrm{d}t'\,R(t-t')f(t') + O(f^2)\,. \tag{2.8.11}$$